

CHENGSHANTOU, China

WMO: 547760

Lat: 37.3897N

Lon: 122.6967E

Elev: 154

StdP: 14.61

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.3	23.1	7.1	8.0	26.9	9.3	8.9	26.7	42.9	26.8	39.6	24.7	23.7	340	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.0	81.7	76.0	80.0	74.9	78.4	74.0	78.1	80.1	76.7	78.5	75.4	77.2	7.6	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.4	143.3	79.5	76.1	137.1	78.1	74.8	131.1	76.7	41.8	80.3	40.4	78.7	39.2	77.2	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
36.8	32.2	28.3	17.6	85.5	4.1	2.1	14.7	87.0	12.3	88.2	10.0	89.4	7.0	91.0		
			15.6	79.3	3.3	1.7	13.1	80.5	11.2	81.5	9.3	82.4	6.9	83.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.6	31.9	33.5	39.6	48.1	57.8	65.3	71.5	75.3	71.3	61.9	49.0	37.2
	DBStd	15.78	5.33	4.92	4.59	4.52	5.13	3.99	3.37	2.56	3.21	5.06	6.80	6.12
	HDD50	1937	561	463	324	89	1	0	0	0	0	2	99	398
	HDD65	4941	1026	883	788	508	230	42	0	0	2	119	480	863
	CDD50	3264	0	0	1	31	244	460	666	783	638	370	70	1
	CDD65	791	0	0	0	0	8	52	202	318	190	21	0	0
	CDH74	2248	0	0	0	0	11	69	498	1380	289	1	0	0
	CDH80	149	0	0	0	0	0	6	45	92	6	0	0	0
Wind	WSAvg	13.2	15.4	14.9	14.5	14.1	12.7	10.9	10.4	10.1	11.3	13.5	15.2	15.7
Precipitation	PrecAvg	29.6	0.5	0.4	0.7	1.8	2.0	3.0	7.8	6.4	3.6	1.5	1.2	0.7
	PrecMax	55.0	2.0	1.5	2.9	5.4	7.2	10.3	36.7	15.1	10.3	4.2	2.8	1.6
	PrecMin	13.5	0.0	0.0	0.0	0.1	0.1	0.3	0.6	2.2	0.1	0.4	0.1	0.1
	PrecStd	9.2	0.4	0.4	0.6	1.5	1.7	2.6	6.6	3.5	3.0	1.1	0.8	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.0	46.6	55.6	64.6	75.2	79.2	84.2	84.8	80.4	73.3	64.8	53.6
		MCWB	40.5	39.6	45.6	49.6	58.6	65.5	77.2	79.1	72.3	65.7	58.1	50.4
	2%	DB	42.9	43.1	50.8	60.0	71.2	75.7	81.0	82.2	78.4	71.1	62.3	50.6
		MCWB	39.4	38.9	43.4	49.1	57.5	65.6	75.4	77.0	72.0	64.5	57.1	46.8
	5%	DB	41.2	41.3	47.6	56.8	68.2	73.2	78.7	80.8	76.9	69.6	60.5	47.9
		MCWB	37.9	38.2	41.4	47.9	56.5	65.9	74.2	76.0	70.7	63.6	55.5	43.2
	10%	DB	39.2	39.8	45.3	54.5	65.7	71.1	76.6	79.4	75.6	68.2	58.5	45.7
		MCWB	35.7	36.9	40.5	47.2	56.1	65.4	73.0	75.4	70.0	62.3	53.3	41.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.6	42.6	48.1	54.2	63.7	70.9	79.2	80.3	76.1	68.5	61.5	51.6
		MCDB	44.0	44.1	52.2	59.1	70.6	73.6	81.7	82.9	77.9	71.3	63.3	53.0
	2%	WB	40.4	40.4	44.8	52.2	61.2	69.5	77.2	78.8	74.3	66.8	58.8	47.5
		MCDB	42.2	41.9	48.3	55.7	65.7	72.1	79.4	80.8	76.2	69.9	61.3	50.1
	5%	WB	38.3	38.9	43.1	50.6	59.8	68.2	75.4	77.6	72.8	65.3	56.3	44.3
		MCDB	40.6	40.5	45.8	53.8	64.0	70.5	77.4	79.5	75.3	68.7	59.5	47.0
	10%	WB	36.3	37.4	41.7	49.2	58.3	67.0	73.9	76.5	71.7	63.5	54.1	41.8
		MCDB	38.9	39.4	44.3	52.3	62.5	69.4	75.6	78.3	74.3	67.4	58.2	45.2
Mean Daily Temperature Range	5% DB	MDBR	6.4	6.5	7.3	7.5	8.3	7.1	6.1	6.0	6.0	6.3	6.9	6.6
		MCDBR	7.9	8.4	11.1	11.9	12.8	10.3	9.0	7.8	7.4	7.0	7.8	8.4
		MCWBR	7.7	7.1	7.5	7.1	6.8	5.9	5.5	4.8	5.2	6.5	7.8	8.4
	5% WB	MCDBR	7.3	7.8	9.5	9.5	10.5	8.1	7.5	6.7	6.3	6.3	7.6	8.1
		MCWBR	8.3	7.1	7.5	7.1	6.8	5.6	5.3	4.6	5.0	6.8	8.6	9.1
Clear-Sky Solar Irradiance	taub	0.462	0.514	0.570	0.603	0.629	0.707	0.693	0.603	0.531	0.553	0.531	0.471	
	taud	1.885	1.801	1.706	1.693	1.688	1.597	1.683	1.865	1.964	1.805	1.803	1.880	
	Ebn at Noon	212	218	220	223	221	204	205	221	229	204	187	197	
	Edh at Noon	47	58	70	75	76	84	76	62	53	57	51	45	
All-Sky Solar Radiation	RadAvg	698	967	1374	1620	1767	1665	1476	1508	1385	1079	747	598	
	RadStd	57	74	107	105	145	154	162	166	107	79	72	38	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page